

Practical programs OS

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Sec: 6

1)VIRTUAL ADDRESS

```
adminrishik@LAPTOP-BSDK1KRC:~$ nano virtualadd.c
adminrishik@LAPTOP-BSDK1KRC:~$ gcc virtualadd.c -o virtualadd
adminrishik@LAPTOP-BSDK1KRC:~$ ./virtualadd
Value of x = 10
Virtual address of x = 0x7ffcfcfb6a44f4
```

2)DEMAND PAGING

```
adminrishik@LAPTOP-BSDK1KRC:~$ nano demandpage.c
adminrishik@LAPTOP-BSDK1KRC:~$ gcc demandpage.c -o demandpage
adminrishik@LAPTOP-BSDK1KRC:~$ ./demandpage
Enter page number (0-3): 2
Page is already in memory.
Physical frame =3
```

3)ADDRESS SPACE

```
adminrishik@LAPTOP-BSDK1KRC:~$ nano addressspace.c
adminrishik@LAPTOP-BSDK1KRC:~$ gcc addressspace.c -o addressspace
adminrishik@LAPTOP-BSDK1KRC:~$ ./addressspace
Code address : 0x5a05924861a9
Data address : 0x5a0592489010
BSS address : 0x5a0592489018
Heap address : 0x5a05a6979010
Stack address : 0x7ffd25bb6ddc
```

4)ADDRESS TRANSILATION

```
adminrishik@LAPTOP-BSDK1KRC:~$ nano adresstransilation.c
adminrishik@LAPTOP-BSDK1KRC:~$ gcc adresstransilation.c -o adresstransilat
ion
adminrishik@LAPTOP-BSDK1KRC:~$ ./adresstransilation
Virtual Address = 250
Page Number = 2
Offset = 50
Frame Number = 1
Physical Address = 150
```

5)DYNAMIC MEMORY

```
adminrishik@LAPTOP-BSDK1KRC:~$ nano dynamicmemo.c
adminrishik@LAPTOP-BSDK1KRC:~$ gcc dynamicmemo.c -o dynamicmemo
adminrishik@LAPTOP-BSDK1KRC:~$ ./dynamicmemo
10 20 30 40 50
```

6)NULL POINTER

```
adminrishik@LAPTOP-BSDK1KRC:~$ nano nullpointer.c
adminrishik@LAPTOP-BSDK1KRC:~$ gcc nullpointer.c -o nullpointer
adminrishik@LAPTOP-BSDK1KRC:~$ ./nullpointer
Before accessing memory
Segmentation fault ./nullpointer
```